

Charles Chen

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SUMMARY

Student programmer and robotics engineer with experience leading VEX Robotics software development, algorithm design, and autonomous systems.

EXPERIENCE

2022 – PRESENT

Smilodon Robotics Programming Lead, Smilodon Robotics, Palo Alto, CA

- Led development of autonomous and driver control software for competitive Vex Robotics robots using C++
- Implemented odometry, 2D Motion Profiling, and Monte Carlo Localization.
- Optimized robot movement accuracy and autonomous consistency during competitions.
- Mentored younger team members in programming and debugging techniques.

2026 Contributor

RW-Template

- Refactored distance sensor reset algorithm to work with non-perpendicular angle offsets.
- Contributed code and testing for open-source robotics software.

SKILLS

- Fluent in C++
- Fluent in HTML and CSS
- Proficient in Python
- Proficient in Java
- Fluent in Chinese
- Advanced Knowledge of Fusion 360

AWARDS AND HONORS

- 2026 Vex Robotics World Championship Think Award
- 2026 Vex Robotics California Region 2 State Finalist
- 2024 FLL Norcal Robot Performance Award
- 2024 FLL 2nd Place Robot Design Award

EDUCATION

EXPECTED 2029

Henry M. Gunn High School, Palo Alto, CA

Clubs: Competitive Programming Club, Cybersecurity Club, Autonomous Robotics Club

